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**CSE 4701 (Principles of Databases)**  
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## **Group 1 - Project 1: Stock Price Trends and Volatility Analysis**

### **1. Identifying Entities and Their Attributes**

#### **1 Stock Entity (stocks table)**

Represents different stocks and their general information.

| Attribute    | Data Type                      | Description                      |
|--------------|--------------------------------|----------------------------------|
| stock_id     | INT PRIMARY KEY                | Database identifier for stock    |
| symbol       | VARCHAR(10) UNIQUE<br>NOT NULL | Stocks symbol e.g. GOOGL, APPL   |
| company_name | VARCHAR(10) UNIQUE<br>NOT NULL | Name of company bound to stock   |
| sector       | VARCHAR(10)                    | General area of business         |
| industry     | VARCHAR(10)                    | Area of business in given sector |

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#### **2 Prices Entity (stocks\_prices table)**

Represents daily information on prices for each stock over a span of three months.

| Attribute | Data Type                         | Description                                     |
|-----------|-----------------------------------|-------------------------------------------------|
| date      | DATE NOT NULL                     | The date in which the price was documented      |
| price_id  | INT PRIMARY KEY<br>AUTO_INCREMENT | Database identifier for instance of stock price |

|                         |                 |                                               |
|-------------------------|-----------------|-----------------------------------------------|
| stock_id (FK to Stocks) | INT             | Database identifier for stock                 |
| open_price              | DECIMAL (10, 2) | Price for the given date when trading opens   |
| close_price             | DECIMAL (10, 2) | Price for the given date when trading closes  |
| high                    | DECIMAL (10, 2) | Peak price for the given date                 |
| low                     | DECIMAL (10, 2) | Lowest price for the given date               |
| volume                  | BIGINT          | Total count of trades made for the given date |

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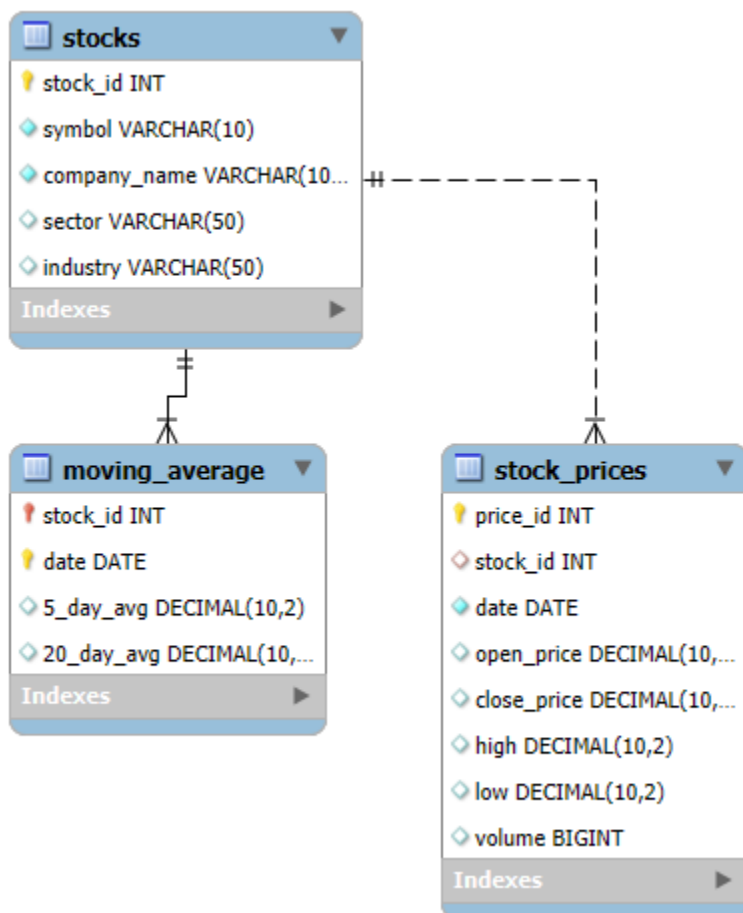
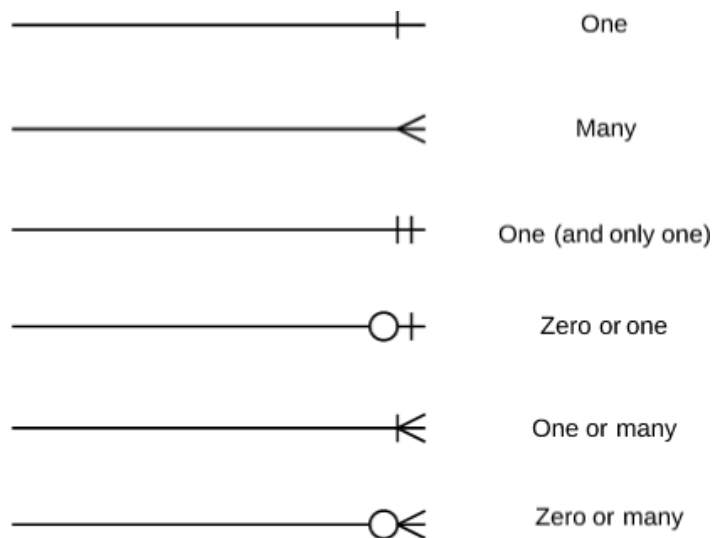
### 3 Averages Entity (moving\_averages table)

Represents both 5-day and 20-day averages for each stock each day.

| Attribute                    | Data Type       | Description                                                           |
|------------------------------|-----------------|-----------------------------------------------------------------------|
| stock_id (PK & FK to Stocks) | INT PRIMARY KEY | Database identifier for stock                                         |
| date                         | DATE NOT NULL   | Latest date among dates of prices for a given stock over 5 or 20 days |
| 5_day_average                | DECIMAL(10, 2)  | The average closing price over 5 days                                 |
| 20_day_average               | DECIMAL(10, 2)  | The average closing price over 20 days                                |

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## 2. Entity Relationship Diagram (ERD)



### 3. Populating Database

#### SQL Schema Creation Script

```
CREATE DATABASE `stockanalysis` /*!40100 DEFAULT CHARACTER SET
utf8mb4 COLLATE utf8mb4_0900_ai_ci */ /*!80016 DEFAULT
ENCRYPTION='N' */;
```

#### 1 Table Creation - stocks table

```
CREATE TABLE stocks (
    stock_id INT PRIMARY KEY AUTO_INCREMENT,
    symbol VARCHAR(10) UNIQUE NOT NULL,
    company_name VARCHAR(10) UNIQUE NOT NULL,
    sector VARCHAR(10),
    industry VARCHAR(10)
);
```

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#### 2 Table Population - stocks table

```
LOAD DATA INFILE ${FILEPATH}
INTO TABLE stocks
FIELDS TERMINATED BY “,”
ENCLOSED BY “”
LINES TERMINATED BY “\n”
IGNORE 1 ROWS;
```

OR

```
INSERT INTO stocks (symbol, company_name, sector, industry) VALUES
('AAPL', 'Apple Inc.', 'Technology', 'Consumer Electronics'),
('MSFT', 'Microsoft Corporation', 'Technology', 'Software'),
('NVDA', 'NVIDIA Corporation', 'Technology', 'Semiconductors'),
('AMZN', 'Amazon.com, Inc.', 'Consumer Discretionary', 'E-Commerce'),
('META', 'Meta Platforms, Inc.', 'Technology', 'Social Media'),
('GOOGL', 'Alphabet Inc. (Class A)', 'Technology', 'Internet Services'),
('TSLA', 'Tesla, Inc.', 'Consumer Discretionary', 'Automobiles'),
('AVGO', 'Broadcom Inc.', 'Technology', 'Semiconductors'),
('GOOG', 'Alphabet Inc. (Class C)', 'Technology', 'Internet Services'),
('BRK-B', 'Berkshire Hathaway Inc. (Class B)', 'Financials', 'Conglomerate');
```

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## 1 Table Creation - stocks\_prices table

```
CREATE TABLE stocks_prices (  
    date DATE NOT NULL,  
    price_id INT PRIMARY KEY AUTO_INCREMENT,  
    stock_id INT,  
    open_price DECIMAL (10, 2),  
    close_price DECIMAL (10, 2),  
    high DECIMAL (10, 2),  
    low DECIMAL (10, 2),  
    volume BIGINT,  
    FOREIGN KEY(stock_id) REFERENCES stocks(stock_id)  
);
```

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## 2 Table Population - stocks\_prices table

```
LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server  
8.0/Uploads/AAPL.csv"  
    INTO TABLE stock_prices  
    FIELDS TERMINATED BY ',' ENCLOSED BY '"'  
    LINES TERMINATED BY '\r\n'  
    IGNORE 1 ROWS  
    (@date, @close_price, @volume, @open_price, @high, @low)  
    SET  
        date = STR_TO_DATE(@date, '%m/%d/%Y'), -- Converts to YYYY-MM-DD  
        close_price = REPLACE(TRIM(@close_price), '$', ''),  
        open_price = REPLACE(TRIM(@open_price), '$', ''),  
        high = REPLACE(TRIM(@high), '$', ''),  
        low = REPLACE(TRIM(@low), '$', ''),  
        volume = TRIM(@volume), -- Removes extra spaces  
        stock_id = 1; -- Adds stock_id
```

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## 1 Table Creation - moving\_average table

```
CREATE TABLE moving_average (
```

```

stock_id INT PRIMARY KEY,
date DATE NOT NULL,
5_day_average DECIMAL (10, 2),
20_day_average DECIMAL (10, 2),
FOREIGN KEY (stock_id) REFERENCES stocks(stock_id)
);

```

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## 2 Table Population - moving\_average table

```

INSERT INTO moving_average (stock_id, DATE, 5_day_avg, 20_day_avg)
SELECT stock_id,
       date AS DATE,
       AVG(close_price) OVER (PARTITION BY stock_id ORDER BY date DESC
ROWS BETWEEN 4 PRECEDING AND CURRENT ROW) AS 5_day_avg,
       AVG(close_price) OVER (PARTITION BY stock_id ORDER BY date DESC
ROWS BETWEEN 19 PRECEDING AND CURRENT ROW) AS 20_day_avg
FROM stock_prices;

```

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## 4. SQL Queries

### 1 The most volatile stocks based on percentage change.

```

SELECT s.stock_id,
       AVG((s.high - s.low) / NULLIF(s.low, 0) * 100) AS pc
FROM stock_prices s
GROUP BY s.stock_id
ORDER BY pc DESC
LIMIT 5;

```

*-- just the top 5 (since there are 10 total)*

|   | stock_id | pc           |
|---|----------|--------------|
| ▶ | 7        | 5.3668039772 |
|   | 3        | 4.4462407197 |
|   | 8        | 4.2995968139 |
|   | 5        | 2.8660318753 |
|   | 6        | 2.3327373511 |

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## 2 Stocks with the highest average trading volume.

```
SELECT s.stock_id,  
       AVG(s.volume) AS tv  
FROM stock_prices s  
GROUP BY s.stock_id  
ORDER BY tv DESC  
LIMIT 5;
```

|   | stock_id | tv             |
|---|----------|----------------|
| ▶ | 3        | 250543168.3333 |
|   | 7        | 83904370.5000  |
|   | 1        | 50659958.8333  |
|   | 4        | 36678194.1667  |
|   | 8        | 34347279.5000  |

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## 3 Stocks that have crossed above or below their moving averages.

```
SELECT  
  s.stock_id,  
  s.date,  
  s.open_price,  
  a.`5_day_avg`  
FROM stock_prices s  
JOIN moving_average a  
  ON s.stock_id = a.stock_id  
  AND s.date = a.date  
WHERE s.open_price > a.`5_day_avg` -- Crosses above  
      OR s.open_price < a.`5_day_avg`; -- Crosses below
```

|   |            |        |        |
|---|------------|--------|--------|
| 4 | 2025-01-23 | 234.10 | 236.18 |
| 4 | 2025-01-24 | 234.50 | 236.03 |
| 4 | 2025-01-27 | 226.21 | 236.59 |
| 4 | 2025-01-28 | 234.29 | 236.99 |
| 4 | 2025-01-29 | 239.02 | 237.77 |
| 4 | 2025-01-30 | 237.14 | 237.59 |
| 4 | 2025-01-31 | 236.50 | 238.43 |
| 4 | 2025-02-03 | 234.06 | 236.73 |
| 4 | 2025-02-04 | 239.01 | 235.87 |
| 4 | 2025-02-05 | 237.02 | 234.01 |
| 4 | 2025-02-06 | 238.01 | 232.56 |
| 4 | 2025-02-07 | 232.50 | 230.87 |
| 4 | 2025-02-10 | 230.55 | 230.78 |
| 4 | 2025-02-11 | 231.92 | 229.48 |
| 4 | 2025-02-12 | 230.46 | 228.25 |
| 4 | 2025-02-13 | 228.85 | 227.04 |
| 4 | 2025-02-14 | 229.20 | 224.28 |
| 4 | 2025-02-18 | 228.82 | 221.09 |
| 4 | 2025-02-19 | 225.52 | 218.32 |
| 4 | 2025-02-20 | 224.78 | 215.86 |
| 4 | 2025-02-21 | 223.28 | 213.04 |
| 4 | 2025-02-24 | 217.45 | 212.18 |
| 4 | 2025-02-25 | 211.63 | 210.64 |
| 4 | 2025-02-26 | 214.94 | 210.10 |
| 4 | 2025-02-27 | 218.35 | 208.68 |
| 4 | 2025-03-03 | 213.35 | 205.02 |
| 5 | 2024-12-03 | 595.00 | 614.74 |
| 5 | 2024-12-04 | 612.96 | 615.87 |
| 5 | 2024-12-05 | 617.08 | 610.65 |

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